Requirement Document: N8N Hands-On Beginner Course

Topic 2: Understanding Workflow Execution & Debugging Basics

1. Document Control

Field	Value
Document Title	Requirement Document: Topic 2 - Understanding Workflow Execution
Course	N8N - Hands On Beginner Course
Target Audience	Trainee / Employee
Version	1.0
Status	Approved
Last Updated	June 20, 2026

2. Objectives & Scope

2.1 Course Objective

The main objective of this learning module is to teach trainees/employees how data moves through a workflow, how to inspect what enters and leaves any given node, and how to identify and debug errors using visual cues, execution history logs, and step-by-step executions.

2.2 Learning Scope

In Scope:

- Inspecting the input data and output data panels of active nodes.
- Simulating a node-level failure using a basic node (e.g. an HTTP Request node configured with an invalid URL).
- Identifying visual indicators for succeeded vs failed nodes (green vs red icons/outlines).
- Performing step-by-step testing on individual nodes ("Test step").
- Navigating and reading the n8n Executions history dashboard.

Out of Scope:

- Advanced error recovery methods (e.g., Error Trigger nodes, "Continue on Fail" routing, Try/Catch blocks).
- Writing code to handle error exceptions.
- Debugging external server/infrastructure crashes.

3. Target Audience Requirements (Pre-requisites)

To successfully complete this hands-on exercise, trainees/employees must have:

- Completed Topic 1: Create Your First Workflow.
- An active n8n editor session.
- Access to the left-hand sidebar navigation (specifically the Executions tab).

4. Functional Requirements

Requirement ID	Section	Requirement Description	Priority
REQ-001	Input/Output Inspection	The trainee shall be able to open any node's configuration panel and toggle between the Input and Output tabs to inspect data schemas.	**High**
REQ-002	Error Simulation	The workflow shall include a node configured to fail intentionally (e.g., an HTTP Request node pointed to a non-existent URL: `https://this-domain-does-not-exist-12345.com`).	**High**
REQ-003	Visual Diagnostics	The system UI shall highlight the failed node in red with a status label stating "Error" or "Failed" after workflow execution.	**High**
REQ-004	Error Data Review	The trainee shall be able to click on the failed node and view the JSON format of the error message under the Output panel.	**High**
REQ-005	Single Step Execution	The editor shall allow executing a single node individually (clicking "Test step") without executing the entire workflow from the trigger.	**High**

Requirement ID	Section	Requirement Description	Priority
REQ-006	Historical Logs	The system shall register every test execution in the Executions sidebar page, listing the execution status, date/time, and duration.	High
REQ-007	Log Inspection	The trainee shall be able to click on a past execution from the history log to inspect the inputs, outputs, and status of that specific run.	Medium

5. Technical Requirements

ID	Technical Requirement	Details
TR-001	Network Connection	The n8n instance must have network access (or simulated DNS failure) to verify connection timeout/invalid DNS errors on the HTTP request node.
TR-002	Execution Save Settings	Ensure "Save execution progress" settings are enabled in n8n configuration so test runs are stored in the execution logs.

6. Stakeholders and Personas

Persona	Role	Need
Trainee / Employee	Student	Needs to learn how to identify where a workflow breaks so they can fix bugs independently.
Instructor / Reviewer	Evaluation	Needs to verify that the student knows how to navigate historical execution logs and inspect nested JSON data payloads.

7. Assumptions & Constraints

- **Assumption 1 (ASM-001):** The n8n instance configuration defaults to saving test runs in the local execution database.
- **Constraint 1 (CON-001):** Historical execution logs might be automatically pruned after 24-48 hours depending on database storage.